Program: **Diploma in Microelectronics**	
Course Code: **6492B**	Course Title: **Basics of Robotics**
Semester: **6**	Credits: **4**
Course Category: **Open Elective**	
Periods per week: **4 (L:4 T:0 P:0)**	Periods per semester: **60**

Course Objectives:

- Learn the types of robots, their anatomy, and key components like actuators and sensors
- Analyze factors such as axes, precision, and capacity to choose appropriate robots for various applications.
- Understand different sensor and actuator types and how to select them for specific robotic tasks.
- Learn basic kinematics principles and use transformation matrices for solving forward and inverse kinematics problems.

Course Prerequisites:

Topic	Course code	Course Title	Semester
Basic Engineering Mathematics Principles		Mathematics I & II	1 & 2
Basic Electrical Principles		Fundamentals of Electrical & Electronics Engineering	2
Sensors		Basic Electronics	2

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive level
CO1	Familiarize with anatomy, specifications and applications of Robots	14	Remembering
CO2	Select suitable sensors and actuators for robotic systems.	14	Understanding
CO3	Identify the right robotic configuration and gripper for specific tasks.	14	Understanding

CO4	Determine the most suitable robotic configuration and gripper for a specific application.	16	Applying
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3	2					
CO2	3	3	3	2	2		
CO3	3	3	3	2	3		
CO4	3	3	3	3	3	1	

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Familiarize with anatomy, specifications and applications of Robots**		
M1.01	Recall the definitions of robots, robotics, and types of robots (manipulators, mobile, aerial).	3	Remembering
M1.02	Describe the anatomy of a robotic manipulator, including links, joints, actuators, and controllers.	5	Understanding
M1.03	Explain kinematic chains, degrees of freedom, and operational considerations for robots.	4	Understanding
M1.04	List applications of robots in various fields like medical, industrial, mining, and space.	2	Remembering

Contents:

Definition, Anatomy, and Applications of Robots
Definitions- Robots, Robotics; Types of Robots- Manipulators, Mobile Robots-wheeled & Legged Robots, Aerial Robots; Anatomy of a robotic manipulator-links, joints, actuators, sensors, controller; open kinematic vs closed kinematic chain; degrees of freedom; Robot considerations for an application- number of axes, work volume, capacity & speed, stroke &reach, Repeatability, Precision and Accuracy, Operating environment, point to point control or continuous path control., Robot Applications- medical, mining, space, defense, security, domestic, entertainment, Industrial Applications-Material handling, welding, Spray painting, Machining.

CO2	Select suitable sensors and actuators for robotic systems.		
M2.01	List the different types of sensors used in robotics, such as touch, force, proximity, and vision sensors.	3	Remembering
M2.02	Explain the difference between internal and external sensors, and describe vision sensor elements.	4	Understanding
M2.03	Describe the classification, working, and advantages/disadvantages of electric, hydraulic, and pneumatic actuators.	4	Understanding
M2.04	Apply knowledge to select suitable sensors and actuators for robotic applications.	3	Applying
	Series Test - I	1	

Contents:

Sensors and Actuators

Sensor classification- touch, force, proximity, vision sensors.
Internal sensors-Position sensors, velocity sensors, acceleration sensors, Force sensors; External sensors-contact type, noncontact type; Vision - Elements of vision sensor, image acquisition, image processing; Selection of sensors.
Actuators for robots- classification-Electric, Hydraulic, Pneumatic actuators; their advantages and disadvantages; Electric actuators- Stepper motors, DC motors, DC servo motors and their drivers, AC motors, Linear actuators, selection of motors; Hydraulic actuators- Components and typical circuit, advantages and disadvantages; Pneumatic Actuators- Components and typical circuit, advantages and disadvantages.

CO3	Identify the right robotic configuration and gripper for specific tasks.		
M3.01	Recall the different robotic configurations (PPP, RPP, RRP, RRR) and their features.	3	Remembering
M3.02	Describe SCARA and PUMA robots and their uses.	3	Understanding
M3.03	Classify types of end effectors such as grippers, tools, and their design considerations.	4	Understanding
M3.04	Choose appropriate configurations and grippers for specific applications.	4	Applying

Contents:

Robotic configurations and end effectors

Robot configurations-PPP, RPP, RRP, RRR; features of SCARA, PUMA Robots; Classification of robots based on motion control methods and drive technologies; 3R concurrent wrist; Classification of End effectors - mechanical grippers, special tools, Magnetic grippers, Vacuum grippers, adhesive grippers, Active and passive grippers, selection and design considerations of grippers in robot.

CO4	**Determine the most suitable robotic configuration and gripper for a specific application.**		
M4.01	Describe basic rotational motions in 2D and 3D space and their composition through sequential applications of rotations.	3	Understanding
M4.02	Explain the representation of rotations using matrices and their properties.	3	Understanding
M4.03	Understand the concept of homogeneous coordinates and transformations for unified representation of positions and orientations.	4	Understanding
M4.04	Apply homogeneous transformation matrices to describe translations, rotations, and scaling in robotic systems.	4	Applying
M4.05	Use forward and inverse kinematics to determine the positions and orientations of robotic arms.	2	Applying
	Series Test - II	1	

Contents:
Kinematics and Motion Planning
Understanding basic rotational motions in 2D and 3D space, Composition of rotations: sequential application of rotations around different axes, Representation of rotations using matrices and their properties, Homogeneous Coordinates and Transformations, Representation of positions and orientations of robotic components in a unified framework. Use of homogeneous transformation matrices to describe translations, rotations, and scaling in a single matrix. Applications in robotic arm manipulation, such as forward and inverse kinematics.

Text /Reference:

T/R	BookTitle/Author
T1	S K Saha ,Introduction to Robotics,, Mc Graw Hill Eduaction
T2	Robert. J. Schilling , "Fundamentals of robotics – Analysis and control", Prentice Hall of India 1996.
T3	R K Mittal and I J Nagrath, "Robotics and Control", Tata McGraw Hill, New Delhi, 2003.
R1	John. J. Craig , Introduction to Robotics (Mechanics and control), Pearson Education Asia 2002.
R2	Ashitava Ghosal, "Robotics-Fundamental concepts and analysis", Oxford University press

Online resources:

Sl.No	Website Link
1	https://www.robotics.org
2	https://www.ros.org/
3	https://www.allaboutcircuits.com/textbook/direct-current/chpt-9/overview-of-sensors/
4	https://www.industrial-robotics.org/
5	https://ocw.mit.edu/courses/mechanical-engineering/2-12-introduction-to-robotics-fall-2007/